

Weekly Report

Items from Last Report:

1. PCB Debugging: Previous PCBs from the lab have been tested and do not give us accurate results. Testing with a small Raspberry Pico with an ADC was performed for a fundamental review of the sensor. Digital output was able to be read from the sensor.

Current Issues:

1. There are no issues.

Goals for Current Week:

1. PCB Debugging: New PCBs should be ordered and shipped so that we can scale up the design. Further testing of the very basic sensor setup can be done to investigate the effectiveness of the sensor.

2. Microcontroller Investigation: As it takes time for new parts to come in. I will investigate microcontrollers that the lab has or microcontrollers to possibly buy to fit the criteria necessary for this project.

Short Term Goal:

1. Hardware MVP: The hardware will be able to read a signal from the pressure sensors as a minimum viable product.

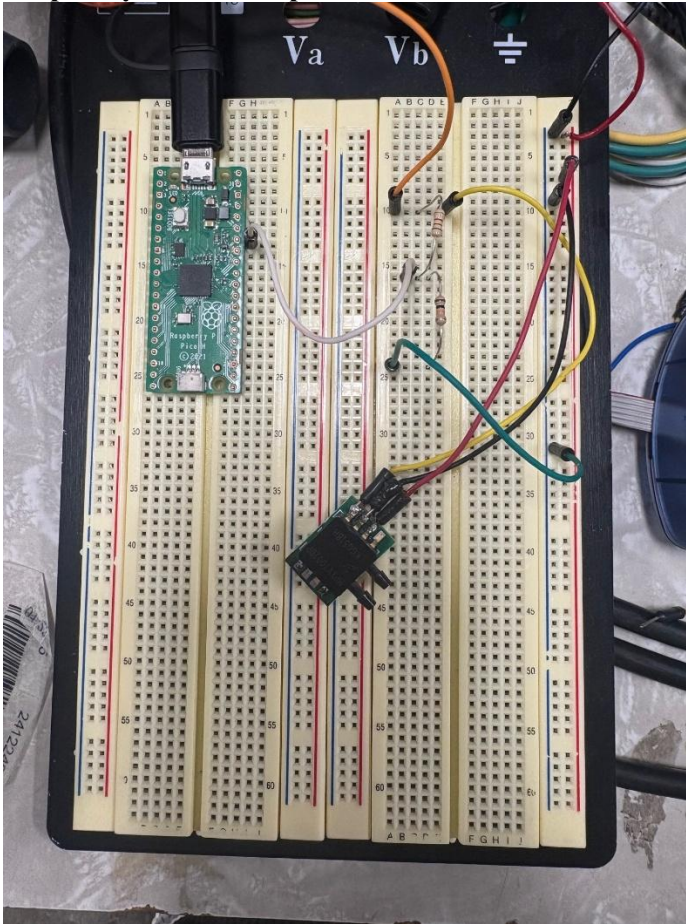
Long Term Goal:

1. Sensor Signal System: The entire system will deliver a functional sensor and decoder setup capable of outputting the clean signal required for fish wake data analysis. July 30, 2026

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Appendix

1. Raspberry Pi Pico setup with differential sensor



2. Digital output and corresponding pressure from value

Monitor Mode	Serial	View Mode	Text	Port	NA - No port selected	Baud rate	115200	Line ending	None
Start Monitoring									
ADC: 2294,	ADC voltage: 1.848 V,	Sensor voltage: 2.800 V,	Pressure: 0.300 kPa						
ADC: 2297,	ADC voltage: 1.851 V,	Sensor voltage: 2.804 V,	Pressure: 0.304 kPa						
ADC: 2296,	ADC voltage: 1.850 V,	Sensor voltage: 2.803 V,	Pressure: 0.303 kPa						
ADC: 2276,	ADC voltage: 1.834 V,	Sensor voltage: 2.778 V,	Pressure: 0.278 kPa						
ADC: 2281,	ADC voltage: 1.838 V,	Sensor voltage: 2.784 V,	Pressure: 0.284 kPa						